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Hiring Manager
Boost inc.

Dear Hiring Manager,

I am writing to express my interest in the Computer Vision Engineer position at Boost inc. Over the past three years I have specialised in developing computer vision solutions for industries where reliability and performance are critical, such as worker safety and oil and gas operations. This experience has given me a strong grounding in **edge AI, model optimisation, and real-time deployment**, all of which are central to the role at Boost inc.

At LivNSense Technologies, I led a team of four engineers to design and deploy deep learning models for real-time **object detection, pose estimation, and tracking** on Intel edge platforms. We reduced false positives by 33 per cent while achieving sub-50 ms inference latency, and optimised inference pipelines with **OpenVINO** and **FP16 quantisation** to ensure deterministic, real-time performance. These achievements required balancing memory, latency, and accuracy under constraints very similar to those faced in retail environments with **occlusion, lighting variations**, and complex consumer behaviours.

Later, at Esbaar, I spearheaded the rollout of **YOLOv5-based detection systems**, improving precision by 15 per cent within three months, and developed a data cleansing pipeline processing 3.6 million data points to accelerate deployment cycles from three days to less than 24 hours. Alongside this, I modernised legacy systems into Kafka-driven microservices with Docker, boosting system resilience and mean time between failures by 35 per cent. These experiences strengthened not only my technical expertise but also my ability to collaborate with cross-functional teams and deliver scalable, production-ready solutions.

I also contributed to building **large-scale data processing pipelines** and improving legacy platforms by introducing **containerisation** and **event-driven architectures**. These changes not only made experimentation faster but also reduced downtime in live systems. I see clear parallels with Boost inc.'s need

for robust, scalable solutions that can operate reliably across both cloud services and edge devices.

While I have not yet worked directly with RKNN toolchains, I have hands-on experience with comparable vendor SDKs including **Intel OpenVINO**, **NVIDIA Jetson**, and **Movidius VPUs**. I am confident in my ability to adapt quickly and become productive with RKNN, just as I have done when mastering other deployment frameworks under demanding project timelines.

What excites me most about this role is the chance to apply my background in high-stakes environments to a consumer-facing domain. The unpredictability of retail settings, with changing packaging, variable lighting, and diverse customer behaviour, mirrors many of the technical challenges I have already tackled. I am confident that my ability to design, refine, and deploy models in such conditions will allow me to contribute effectively from day one.

I would welcome the opportunity to discuss how my experience can support Boost Inc.'s mission to transform unattended retail. Thank you for considering my application.

Yours sincerely,
Dexter Fernandes